

RVM & PecoDrop Hardware Diagnostics

Implementation and troubleshooting guide

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Architecture: The public RVM and enterprise PecoDrop remain independent applications with separate firmware and protocols. Admin tests reuse the kiosk's active serial connection; they never open a competing COM port.

Safety rules

- Manual actuator tests stop normal intake.
- RVM gates auto-close after three seconds. Only one gate is opened at a time.
- PecoDrop retains its five-second host lease, close/detach behavior, and `SERVO:ALL:CLOSE` shutdown.
- Closing Admin returns hardware to a safe state.
- Servo status means *commanded* position. Physical position is not verified because no limit-switch feedback is installed.

Public RVM — General Public

Component	Pin/interface	Diagnostic output
Bottom / middle / top optical sensors	D2 / D7 / D8	Raw digital state
Chamber ultrasonic	Trig D4, Echo D3	Distance cm and valid/no-echo
Entrance ultrasonic	Trig D12, Echo D13	Distance cm and valid/no-echo
Metal / UBC sensors	D5 / D10	Raw classification inputs
Bin sensor	D11	Raw and interpreted blocked state
Main / entrance servos	D9 / D6	Guarded open, close, automatic closure

RVM commands

Command	Purpose
<code>PROTO?</code>	Firmware identity and capabilities
<code>DIAG:SNAPSHOT</code>	All eight sensors, both gates, calibration and uptime
<code>DIAG:SELF_TEST</code>	Safe sensors-only snapshot
<code>DIAG:ENTER</code> / <code>DIAG:EXIT</code>	Stop intake and close both gates
<code>DIAG:GATE:OPEN CLOSE</code>	Main gate test
<code>DIAG:ENTRANCE:OPEN CLOSE</code>	Entrance gate test

Bench confirmation: D11 polarity, D10 sensor type, and the mechanically correct main-gate open angle should be confirmed on the physical RVM. Existing firmware behavior was preserved.

PecoDrop — Enterprise / Corporate

Dual landscape displays, optical QR scanner/keypad,  plastic,  cans/metal, and  weighed paper remain the canonical configuration.

Stream	Sensors	Actuators
 Plastic	Entrance + three sizing ultrasonics; bin sensor	Iris and drop gate
 Metal/cans	Entrance + three sizing ultrasonics; inductive sensor; bin sensor	Iris and drop gate
 Paper	Top + bottom ultrasonics; HX711 load cell; bin sensor	Iris and drop gate
Safety	MQ-6 digital alarm, calibration and host-lease state	All-close and PWM detach

PecoDrop commands

Command	Purpose
PROTO?	PecoDrop identity and capabilities
DIAG:SNAPSHOT	Ten ultrasonic readings, metal, three bins, MQ-6, HX711, lease and uptime
DIAG:SELF_TEST	Safe sensor snapshot with intake stopped
SCALE	HX711 raw, tare and signed grams
CALIBRATE	Full authoritative calibration
SERVO: . . . :OPEN CLOSE	Existing six individual actuator tests
SERVO:ALL:CLOSE	Emergency safe-close

Admin workflow

1. Open Admin while the kiosk owns the Arduino connection.
2. Confirm the green connected status.
3. Select **Refresh All** and inspect timestamped RX output.
4. Activate each sensor physically and refresh again; compare raw and interpreted states.
5. Run calibration only with all apertures/chambers empty.
6. Run one actuator at a time while personnel remain clear.
7. Use CLOSE ALL / Safe Stop before leaving Admin.
8. Save the console text with the machine ID and firmware version in the maintenance record.

Acceptance checklist

- Correct firmware identity; no wrong-app protocol.
- No second COM connection is created.
- Every sensor changes state or returns a valid distance.
- Every commanded servo moves in the correct direction and safely closes.
- Disconnecting USB or expiring the PecoDrop lease closes actuators.
- Normal PLASTIC/CAN/UBC/REJECT flow works after RVM diagnostics.
- Normal PecoDrop circle/triangle/square workflow works after diagnostics.

Build verification: RVMDesktopApp and PecoDropDesktopApp compiled with .NET 8 (0 warnings, 0 errors). Arduino Uno firmware compiled at 33% flash / 57% SRAM. PecoDrop Mega firmware compiled at 8% flash / 15% SRAM.